THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 5: **Number Play** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 3465	alternate digit sums 9 and 9 differ by 0, a multiple of 11
2	**(d)** 4518	even and digit sum 18 is a multiple of 3
3	**(a)** 6174	Kaprekar routine: arrange digits in descending and ascending order and subtract
4	**(c)** 39	Sum $= 3\times$ middle number
5	**(b)** 80	Sum $= 5\times$ middle number
6	**(c)** 14	$(7+5)\times 2-10=2\times 7$
7	**(a)** 1	Check the rule for 9
8	**(b)** odd	Odd + odd = even, even + odd = odd
9	**(d)** 8239	alternate digit sums 11 and 11 differ by 0, a multiple of 11
10	**(a)** 6150	even and digit sum 12 is a multiple of 3

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Sum of digits at odd places and even places: 13 and 13 (1); difference 0, so yes (1).	1+1
12	Largest arrangement 6532, smallest 2356 (1); difference 4176 (1).	1+1
13	$(13+6)\times 3=57$; minus 18 gives 39 (1); result $=3\times 13$ (1).	1+1
14	Sum of digits at odd places and even places: 17 and 6 (1); difference 11, so yes (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	By 8: Yes, last three digits 640 are divisible by 8 (1); by 11: Yes, alternate digit sums 9 and 9 differ by 0, a multiple of 11 (1); conclusion (1).	1+1+1
16	Steps: 5321-1235=4086; 8640-0468=8172; 8721-1278=7443; 7443-3447=3996; 9963-3699=6264; 6642-2466=4176; 7641-1467=6174 (2); reaches 6174 after 7 steps (1).	2+1
17	47+74=121 (1); 121$=11\times 11$ (1); in general $10a+b+10b+a=11(a+b)$ (1).	1+1+1
	OR: By 8: Yes, last three digits 128 are divisible by 8 (1); by 11: Yes, alternate digit sums 16 and 5 differ by 11, a multiple of 11 (1); conclusion (1).	1+1+1
18	Steps: 8740-0478=8262; 8622-2268=6354; 6543-3456=3087; 8730-0378=8352; 8532-2358=6174 (2); reaches 6174 after 5 steps (1).	2+1
	OR: 63+36=99 (1); 99$=11\times 9$ (1); in general $10a+b+10b+a=11(a+b)$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Conditions of 5 and 9 on the stars (2); numbers: 4230, 4635 (2); verify one (1).	2+2+1
	OR: $83-38=45=9\times 5$ (2); in general $(10a+b)-(10b+a)=9(a-b)$ (2); hence divisible by 9 (1).	2+2+1
20	Conditions of 10 and 9 on the stars (2); numbers: 6750 (2); verify one (1).	2+2+1
	OR: $61-16=45=9\times 5$ (2); in general $(10a+b)-(10b+a)=9(a-b)$ (2); hence divisible by 9 (1).	2+2+1

Mistakes to avoid:

- Testing 11 with the digit *sum*; the test uses the **difference** of the sums at odd and even places.
- Saying a number is divisible by 6 because it is divisible by 2 or 3; it must be divisible by **both**.
- Thinking that divisibility by 4 needs the digit sum; look at the last **two** digits (last **three** for 8).
- Subtracting in the wrong order in Kaprekar's routine: always take (largest arrangement) − (smallest arrangement).
- Ignoring the leading zero in the smallest arrangement (e.g. 0378 is 378), which gives a wrong difference.
- Assuming a number divisible by 4 and 6 must be divisible by 24; check with 12 (divisible by both, not by 24).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 2

Chapter 5: **Number Play** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 6174	Kaprekar routine: arrange digits in descending and ascending order and subtract
2	**(d)** 78	Sum $= 3\times$ middle number
3	**(a)** 155	Sum $= 5\times$ middle number
4	**(d)** 27	$(9+4)\times 3-12=3\times 9$
5	**(c)** 0	Check the rule for 3
6	**(c)** even	One of them is even
7	**(b)** 5214	alternate digit sums 6 and 6 differ by 0, a multiple of 11
8	**(a)** 7296	even and digit sum 24 is a multiple of 3
9	**(d)** 9171	Kaprekar routine: arrange digits in descending and ascending order and subtract
10	**(c)** 114	Sum $= 3\times$ middle number

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Largest arrangement 8741, smallest 1478 (1); difference 7263 (1).	1+1
12	$(21+4)\times 5=125$; minus 20 gives 105 (1); result $=5\times 21$ (1).	1+1
13	Sum of digits at odd places and even places: 3 and 14 (1); difference 11, so yes (1).	1+1
14	Largest arrangement 9210, smallest 0129 (1); difference 9081 (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	By 8: Yes, last three digits 896 are divisible by 8 (1); by 11: Yes, alternate digit sums 18 and 18 differ by 0, a multiple of 11 (1); conclusion (1).	1+1+1
16	Steps: 9632-2369=7263; 7632-2367=5265; 6552-2556=3996; 9963-3699=6264; 6642-2466=4176; 7641-1467=6174 (2); reaches 6174 after 6 steps (1).	2+1
17	29+92=121 (1); $121=11\times 11$ (1); in general $10a+b+10b+a=11(a+b)$ (1).	1+1+1
	OR: By 8: Yes, last three digits 728 are divisible by 8 (1); by 11: No, alternate digit sums 24 and 3 differ by 21, not a multiple of 11 (1); conclusion (1).	1+1+1
18	Steps: 8510-0158=8352; 8532-2358=6174 (2); reaches 6174 after 2 steps (1).	2+1
	OR: 85+58=143 (1); $143=11\times 13$ (1); in general $10a+b+10b+a=11(a+b)$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Conditions of 10 and 9 on the stars (2); numbers: 2340 (2); verify one (1).	2+2+1
	OR: $94-49=45=9\times 5$ (2); in general $(10a+b)-(10b+a)=9(a-b)$ (2); hence divisible by 9 (1).	2+2+1
20	Conditions of 5 and 9 on the stars (2); numbers: 8010, 8415, 8910 (2); verify one (1).	2+2+1
	OR: $52-25=27=9\times 3$ (2); in general $(10a+b)-(10b+a)=9(a-b)$ (2); hence divisible by 9 (1).	2+2+1

Mistakes to avoid:

- Testing 11 with the digit *sum*; the test uses the **difference** of the sums at odd and even places.
- Saying a number is divisible by 6 because it is divisible by 2 or 3; it must be divisible by **both**.
- Thinking that divisibility by 4 needs the digit sum; look at the last **two** digits (last **three** for 8).
- Subtracting in the wrong order in Kaprekar's routine: always take (largest arrangement) − (smallest arrangement).
- Ignoring the leading zero in the smallest arrangement (e.g. 0378 is 378), which gives a wrong difference.
- Assuming a number divisible by 4 and 6 must be divisible by 24; check with 12 (divisible by both, not by 24).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 3

Chapter 5: **Number Play** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 175	Sum $= 5\times$ middle number
2	**(a)** 48	$(12+6)\times 4-24=4\times 12$
3	**(d)** 4	Check the rule for 9
4	**(b)** 12	4 and 3 have no common factor
5	**(c)** 7183	alternate digit sums 4 and 15 differ by 11, a multiple of 11
6	**(a)** 5322	even and digit sum 12 is a multiple of 3
7	**(c)** 7173	Kaprekar routine: arrange digits in descending and ascending order and subtract
8	**(c)** 147	Sum $= 3\times$ middle number
9	**(a)** 210	Sum $= 5\times$ middle number
10	**(b)** 75	$(15+3)\times 5-15=5\times 15$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$(17+7)\times 2=48$; minus 14 gives 34 (1); result $=2\times 17$ (1).	1+1
12	Sum of digits at odd places and even places: 9 and 8 (1); difference 1, so no (1).	1+1
13	Largest arrangement 5430, smallest 0345 (1); difference 5085 (1).	1+1
14	$(26+3)\times 4=116$; minus 12 gives 104 (1); result $=4\times 26$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	By 8: Yes, last three digits 624 are divisible by 8 (1); by 11: Yes, alternate digit sums 15 and 4 differ by 11, a multiple of 11 (1); conclusion (1).	1+1+1
16	Steps: 8621-1268=7353; 7533-3357=4176; 7641-1467=6174 (2); reaches 6174 after 3 steps (1).	2+1
17	38+83=121 (1); $121=11\times 11$ (1); in general $10a+b+10b+a=11(a+b)$ (1).	1+1+1
	OR: By 8: No, last three digits 652 are not divisible by 8 (1); by 11: Yes, alternate digit sums 16 and 5 differ by 11, a multiple of 11 (1); conclusion (1).	1+1+1
18	Steps: 9430-0349=9081; 9810-0189=9621; 9621-1269=8352; 8532-2358=6174 (2); reaches 6174 after 4 steps (1).	2+1
	OR: 74+47=121 (1); $121=11\times 11$ (1); in general $10a+b+10b+a=11(a+b)$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Conditions of 10 and 3 on the stars (2); numbers: 3270, 3570, 3870 (2); verify one (1).	2+2+1
	OR: $71-17=54=9\times 6$ (2); in general $(10a+b)-(10b+a)=9(a-b)$ (2); hence divisible by 9 (1).	2+2+1
20	Conditions of 10 and 9 on the stars (2); numbers: 7560 (2); verify one (1).	2+2+1
	OR: $85-58=27=9\times 3$ (2); in general $(10a+b)-(10b+a)=9(a-b)$ (2); hence divisible by 9 (1).	2+2+1

Mistakes to avoid:

- Testing 11 with the digit *sum*; the test uses the **difference** of the sums at odd and even places.
- Saying a number is divisible by 6 because it is divisible by 2 or 3; it must be divisible by **both**.
- Thinking that divisibility by 4 needs the digit sum; look at the last **two** digits (last **three** for 8).
- Subtracting in the wrong order in Kaprekar's routine: always take (largest arrangement) − (smallest arrangement).
- Ignoring the leading zero in the smallest arrangement (e.g. 0378 is 378), which gives a wrong difference.
- Assuming a number divisible by 4 and 6 must be divisible by 24; check with 12 (divisible by both, not by 24).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.